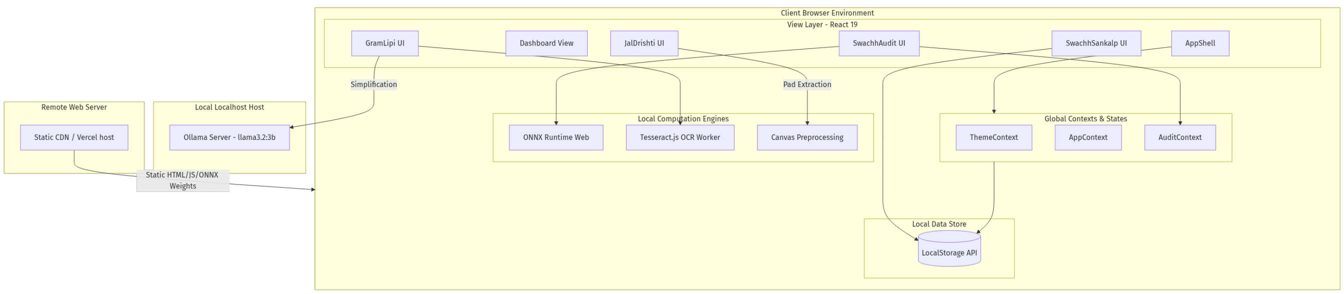
GramSetu AI - Software Architecture Specification

This document details the architectural layout, modules separation, global contexts, hooks pipelines, and data flow of GramSetu AI.

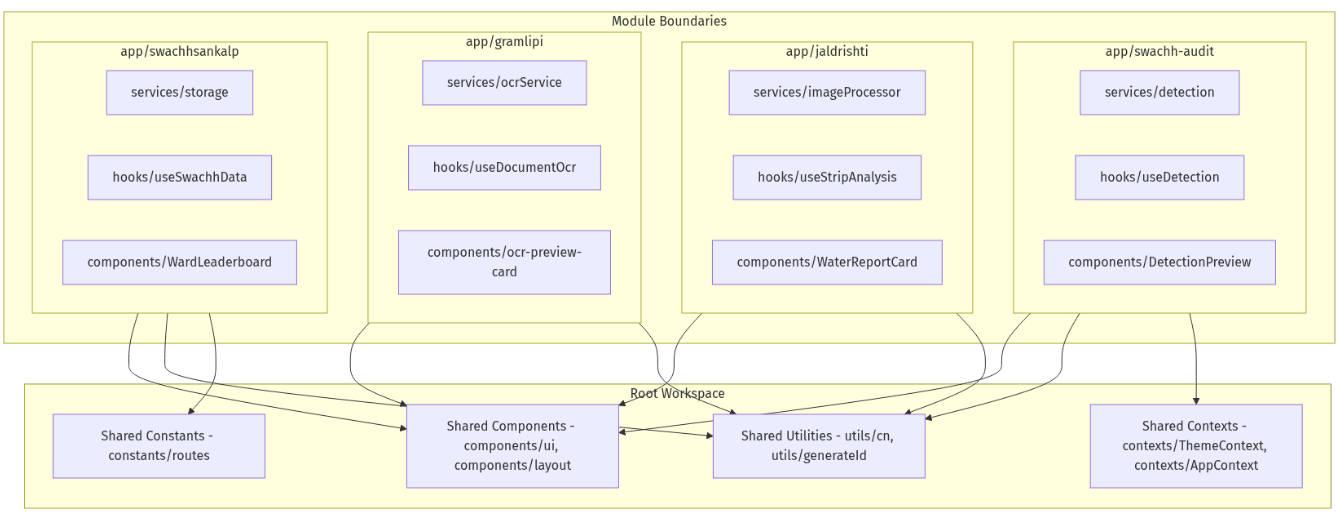
1. Overall System Architecture

GramSetu AI utilizes a local-first, zero-cloud architecture. It runs in-browser utilizing client hardware acceleration.



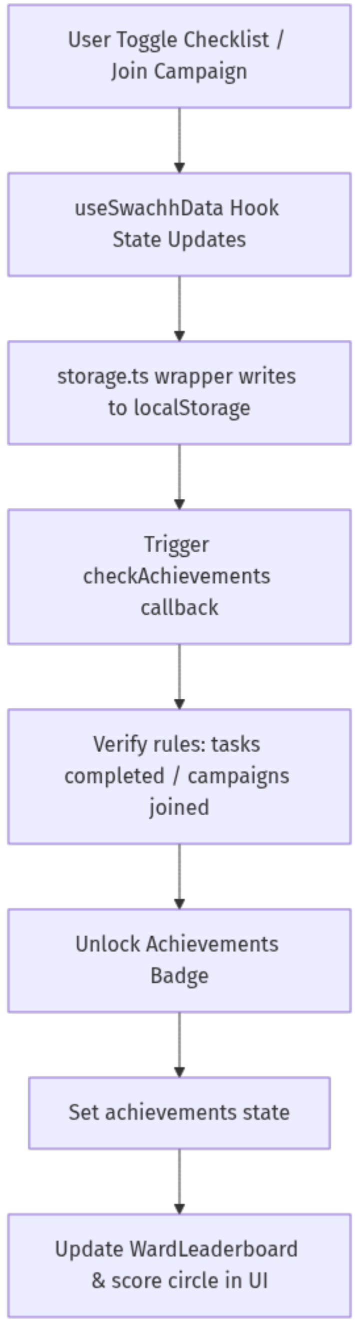
2. Module Sandboxing & Interactions

Each module inside the **`app/`** directory (such as **`swachh-audit`**, **`jaldrishti`**, **`gramlipi`**, and **`swachhsankalp`**) is structured as an isolated sandbox containing its own components, services, and hooks. Shared layouts and core utility functions reside at the root.



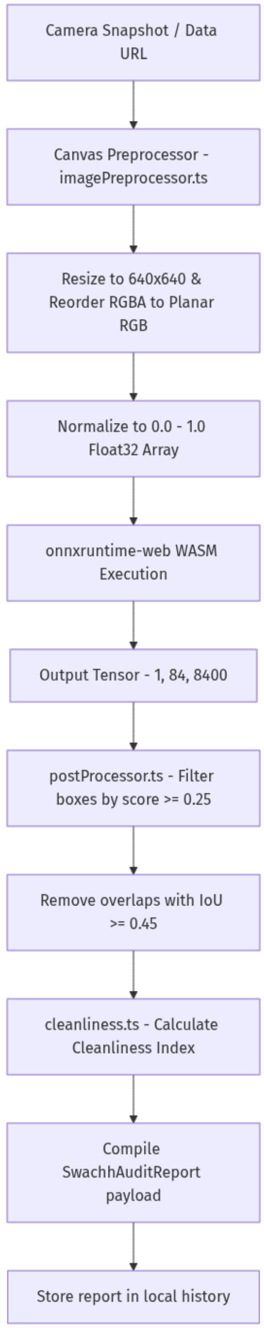
3. Data Flow & Local State Synchronization

Data persistence uses browser **`localStorage`**. Changes in UI components update the local React state, which is synchronously written to localStorage via hook wrappers.



4. AI Inference Execution Flow (SwachhAudit)

The pipeline for processing garbage audits starts at camera capture, flows through scale/normalization canvas layers, runs through on-device models, and filters outputs using Non-Maximum Suppression (NMS).



5. UI Component Hierarchy

The Root Layout renders the shell (**`AppShell`**) which houses navigation links, routing indicators, and the main page router.

```
graph TD
    RootLayout[Root Layout - layout.tsx]
    --> AppShell[AppShell.tsx]

    AppShell --> TopNavbar[TopNavbar.tsx]
    AppShell --> Sidebar[Sidebar.tsx]
    AppShell --> MobileNavigation[MobileNavigation.tsx]
    AppShell --> MainContent[Main Content Router]
```

```
    TopNavbar --> Breadcrumb[Breadcrumb.tsx]
    TopNavbar --> ThemeToggle[ThemeToggle / Profile]

    MainContent --> Dashboard[Dashboard - app/dashboard/page.tsx]
    MainContent --> SwachhAuditPage[SwachhAudit - app/swachh-audit/page.tsx]
    MainContent --> JalDrishtiPage[JalDrishti - app/jaldrishti/page.tsx]
    MainContent --> GramLipiPage[GramLipi - app/gramlipi/page.tsx]
    MainContent --> SwachhSankalpPage[SwachhSankalp - app/swachhsankalp/page.tsx]
```

6. Folder Dependency Flow

Dependencies flow from modular directories to root shared layers. Modules do not import from other modules.